MILITARY SPECIFICATION

SWITCH, ANTI-DISTURBANCE FOR M56 MINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the inspection and test for the Anti-disturbance switch used in the M56 Anti-tank mine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification for

STANDARDS

MILITARY

M1L-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-202	- Test Methods for Electronic and Electrical, Component Parts
MIL-STD-883	- Test Methods and Procedures for Microelectronics
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment for Inert Ammunition Components

FSC: 1345

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

PRODUCT DRAWING

9275347 - Disturbance Switch Assembly

INSPECTION EQUIPMENT DRAWING

9278643 - AD Switch Test Circuit

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. - The assembly shall comply with all requirements specified on Drawing (dwg.) 9275347 and with all requirements specified in applicable specifications.

3.3 Functioning. - The switch shall function at some instance when it is moved a maximum of three times (3 x's) through an angle of sixty degrees (600), i.e., thirty degrees (30°) from both sides of the vertical.

3.4 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner, They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning agent or method used shall not be injurious to any part, nor shall the part be contaminated by the cleaning agents.

3.5 Submission of Results of Contractor-conducted Examinations and Tests. - The contractor shall forward copies of the results as required by the contract. Results shall be submitted within thirty (30) days at the completion of each lot tested (see 6.4).

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3.6 First Article Inspection. - This specification makes provisions for first article inspection. Requirement for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. -At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority

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will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. -Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of five (5) sets of parts and twenty five (25) complete switch assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as, to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. -Assemblies and components will be subjected by the Government to any or all of the examinations and tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any component or subassembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an inert component or subassembly to comply with any of the stated requirements.

4.3 Inspection provisions

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4.3.1 Lot formation. -The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. -The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Disturbance Switch Assembly (see dwg. 9275347).

Categories	Defects	Method of Inspection	Code No. (see 6.2)

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Critical: None defined.

Major:		AQL 0.40 percent	
101.	Length of lead(s), minimum (min.)	Gage	01001
102.	Improper lead bending	Gage	01002
Minor:		AQL 0.65 percent	
201.	Crimp(s) missing	Visual	01003
202.	Evidence of poor workmanship (see 3.4)	Visual	01004

4.3.3 Testing

4.3.3.1 Plating of Parts (see dwg. 9275347).

4.3.3.1.1 Ball-- Major Defect - Code No. 02001.-Five (5) balls shall be randomly selected from each plating batch and examined for the applicable requirement (see dwg. 9275347) as specified in 4.4.1. The lot shall be rejected if one (1) or more balls are classed defective. (Destructive Test)

4.3.3.1.2 Header - Major Defect - Code No. 03001.-Five (5) headers shall be randomly selected from each plating batch and examined for the applicable requirements (see dwg. 9275347) as specified in 4.4.2. The lot shall be rejected if one (1) or more headers are classed defective. (Destructive Test)

4.3.3.1.3 Cap - Major Defect - Code No. 04001.-Five (5) caps shall be randomly selected from each plating batch and examined for the applicable requirements (see dwg. 9275347) as specified in 4.4.3. The lot shall be rejected if one (1) or more headers are classed defective. (Destructive Test)

4.3.3.2 Insulation Resistance (see dwg. 9275347). - Major Defect. - Code No. 01005 - Each welded lot of switches shall be tested for the applicable requirements in accordance with 4.4.4. The sampling plan shall be in accordance with M1L-STD-105, Level II using an AQL of 0.40 percent (Non-Destructive Test).

4.3.3.3 Gross Leakage (see dwg. 9275347). - Major Defect - Code NO. 01006. - Each welded lot of switches shall be tested for the applicable requirements in accordance with Method 1014, Test Condition C, Step 1 of MIL-STD-883. Sampling shall be in accordance with M1L-STD-105, level II using an AQL of 0.40 percent (Non-Destructive Test).

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4.3.3.4 Fine Leakage (see dwg. 9275347). - Major Defect - Code No. 01007. - Each welded lot of switches shall be tested for the applicable requirements in accordance with Test Method 112, Condition C of MIL-STD-202. Sampling shall be in accordance with MIL-STD-105, level II, using an AQL of 0.110 percent. (Non-Destructive test)

4.3.3.5 Functioning Test (see 3.3).- Major Defect - Code No. 01008. - Each welded lot shall be tested 100 percent for the specified requirements in accordance with 4.4.5. Any switch classed defective shall be removed from the lot. (Non-Destructive test)

4.3.4 Inspection Equipment. - Inspection equipment required to perform the inspection prescribed in this specification is identified in the examination (4.3.2) paragraph herein. See 6.3 for details concerning responsibilities for inspection equipment design and approval.

4.4 TEST METHODS AND PROCEDURES

4.4.1 Plating of disturbance switch ball. - The test speciman shall be cross-sectioned through the major diameter of the sample, polished and examined microscopically to measure the plating thickness. The sample shall be classed defective if it fails to comply with all of the plating requirements. Testing is destructive and the sample shall not be returned to the lot.

4.4.2 Plating of disturbance switch header. The test speciman shall be cross-sectioned across the diameter and through the center pin, polished and examined microscopically to measure the plating thickness. The sample shall be classed defective if it fails to comply with all of the plating requirements. Testing is destructive and the sample shall not be returned to the lot.

4.4.3 Plating of disturbance switch cap. The test speciman shall be cross-sectioned through the center of the cap, polished and examined microscopically to measure the plating thickness. The sample shall be classed defective if it fails to comply with all of the plating requirements. Testing is destructive and the sample shall not be returned to the lot.

4.4.4 Insulation Resistance. - The switch shall be tested for the required insulation using a megohmmeter producing a minimum voltage output of the specified required potential for

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the minimum time specified on the applicable drawing. Samples failing to comply with all of the specified requirements shall be classed defective. Testing is non-destructive and accepted samples shall be returned to the lot.

4.4.5 Functioning. - The switch shall be functioned in a Government approved fixture that imparts the required motion and utilizes the circuit on drawing 9278643. Any switch failing to comply with all of the specified requirements shall be classed defective. Testing is non-destructive and only accepted switches shall be returned to the lot.

5. PREPARATIONS FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 Level C. - The switches shall be packaged in accordance with MIL-STD-1169.

5.2 Packing

5.2.1 Level C. - The switches shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. - Marking of the carton shall comply with MIL-sTD-1169.

5.4 Shipping. - When switches of more than one lot are shipped together each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data. -Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future

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data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs .-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in M1L-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. -Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists

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(DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of results of contractor conducted examinations and tests (see 3.5). - The contractor shall send his results to:

Commander
Picatinny Arsenal
ATTN : SARPA-QA-A-M
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1345-A-042

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